Energy transfer in chemical reactions

Learning outcomes

By the end of this section you should be able to:

- Describe the law of conservation of energy and its application to chemical reactions.
- Identify system and surroundings and direction of energy transfer for a chemical reaction.
- Explain the difference between heat and temperature.

Fire has been an important tool for human survival and development throughout history and has been an important part of many cultures and religions. Despite its significance, fire was considered quite mysterious throughout much of history. This is because it was always viewed as a physical substance or entity. Instead, fire is better described as an event or process that releases energy in the form of heat, sound and light. It is a chemical reaction that results in this process when a fuel (such as wood) reacts with oxygen in the presence of an ignition source.

A fire triangle, as shown in **Figure 1**, is often used to display the three components necessary for fire. The triangle can be simple to represent a small fire or can be scaled up to consider fire spread over landscapes. It is important for us to consider these larger scale fire triangles as global climates change, influencing more factors involved in the 'wildfire' and 'fire regime' triangles.

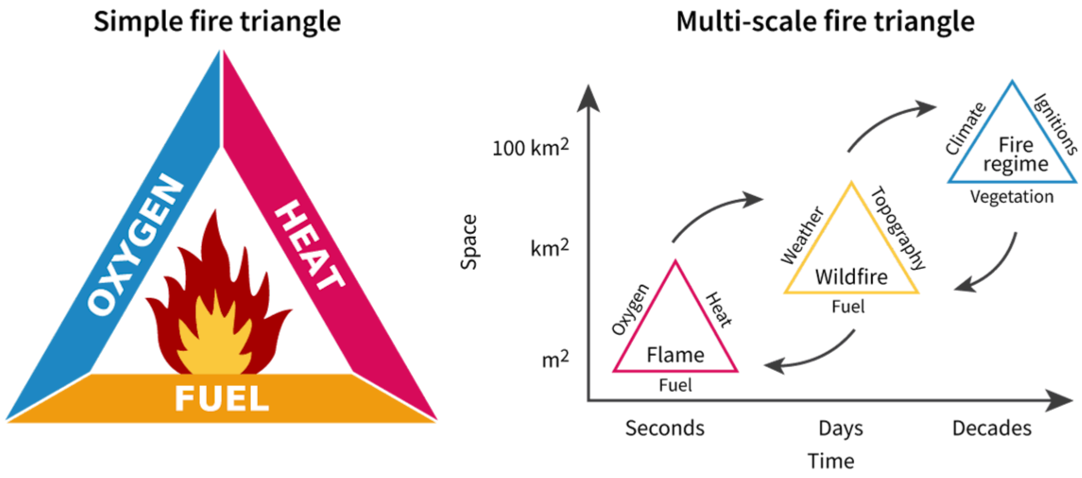

Figure 1. Simple and multi-scale fire triangles.

More information for figure 1

The image depicts two concepts related to fire dynamics. On the left is a classic fire triangle with three sides labeled 'Oxygen', 'Heat', and 'Fuel', with a flame symbolizing fire in the center. On the right, there is a multi-scale representation of the fire triangle concept, shown as a series of arrows pointing from a smaller red triangle to a medium yellow triangle and finally a larger blue triangle, indicating the progression or scaling of fire over landscapes.

[Generated by AI]

But how does a chemical reaction from fuel make a fire? How does it produce heat? Why does the temperature of the environment around a fire increase? What is the difference between heat and temperature?

What is energy?

Energy is an important term in chemistry. The term energy used with reference to energy levels that electrons occupy is covered in section S1.3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/), and ionisation energy considered as a quantitative measurement of the ease to which an electron can be removed from an atom is described in section S3.1.3b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/). But what exactly is energy? How is it measured and what units does it have?

Energy is the ability to do work or cause a change and is measured in units of joules (J). Energy exists in many different forms. **Figure 2** illustrates some different energy forms categorised into energy from movement, stored energy and energy we can experience.

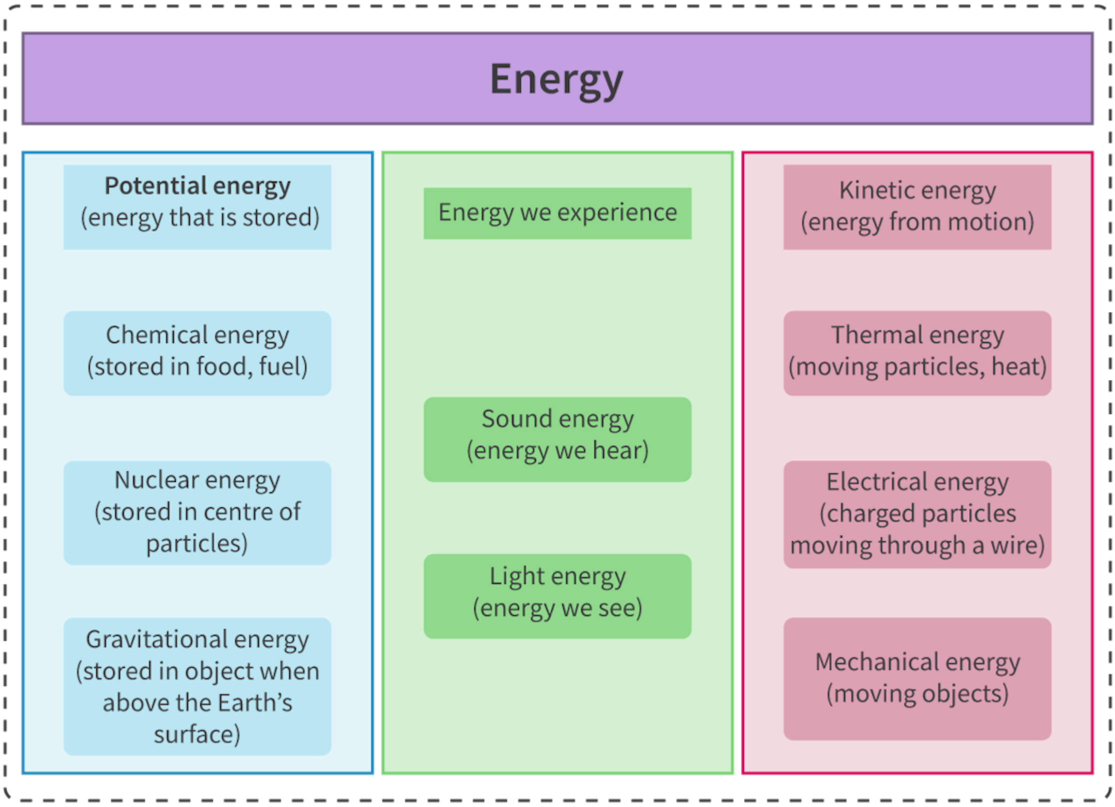

Figure 2. Different forms of energy.

More information for figure 2

The diagram categorizes different forms of energy. It is divided into three columns.

1. **Potential Energy (energy that is stored):**
2. **Chemical energy:** Stored in food and fuel.
3. **Nuclear energy:** Stored in the center of particles.
4. **Gravitational energy:** Stored in an object when above the Earth's surface.

5. **Energy We Experience:**

6. **Sound energy:** Energy we hear.
7. **Light energy:** Energy we see.

8. **Kinetic Energy (energy from motion):**

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9. **Thermal energy:** Moving particles, heat.
10. **Electrical energy:** Charged particles moving through a wire.
11. **Mechanical energy:** Moving objects.

The diagram visually shows the relationship between different energy types, categorizing them into stored, experienced, and motion-related energy.

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Whereas energy is a term most of us are familiar with, it is a strange concept to define on its own. More commonly, we are familiar with the transfer of energy. The three main processes for energy transfer include:

1. Conduction: transfer of energy by direct contact
2. Convection: transfer of energy by the movement of fluids
3. Radiation: transfer of energy by electromagnetic waves.

These three processes for energy transfer are illustrated in **Figure 3** using a pot of water boiling as an example to show how thermal energy can be transferred.

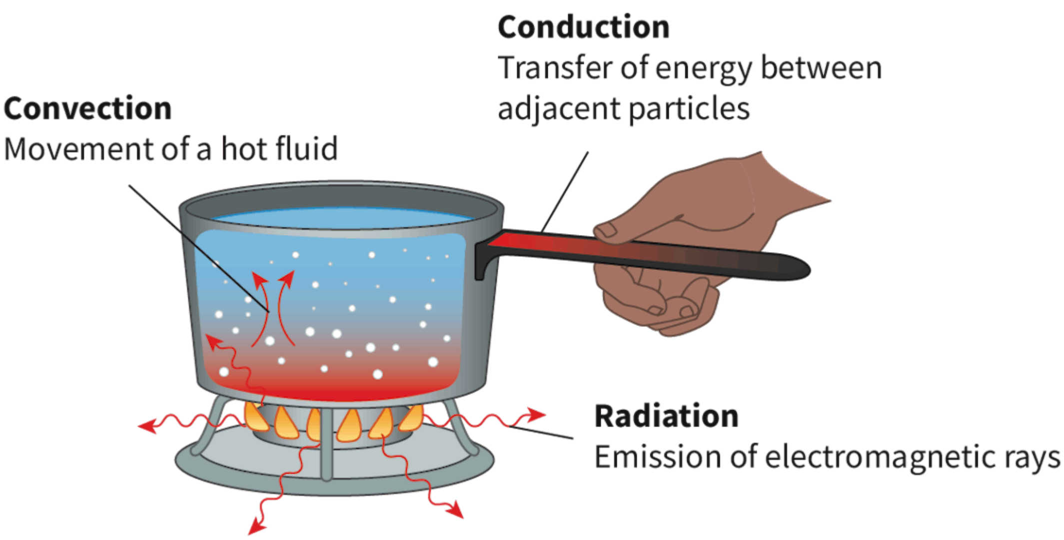

Figure 3. Transfer of thermal energy through conduction, convection and radiation.

More information for figure 3

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The diagram illustrates the transfer of thermal energy using a boiling pot of water. It highlights three methods: conduction, convection, and radiation. Conduction is shown by the hand holding the pot handle, indicating energy transfer between adjacent particles, like from the pot to the hand. Convection is depicted by the movement of hot water within the pot, represented by arrows rising through the water, suggesting the movement of hot fluid upwards. Radiation is illustrated by wavy arrows emanating from the flame below the pot, signifying the emission of electromagnetic rays heating the pot from beneath.

[Generated by AI]

When a transfer of energy occurs, energy may change from one form to another, but, the total amount of energy will always remain the same in a closed system. This is known as the law of conservation of energy. For example, when we turn on a light bulb, energy is transformed from electrical energy into light, thermal and some sound energy. As a society we place emphasis on the energy transferred to the desired energy form (light) and rate efficiency of materials on that. This rating of efficiency makes it seem like energy is lost, but it is not. It has also been transformed into less useful forms. The total amount of energy remains constant.

Physical changes and energy transfer

Section S1.1.2a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/) covers physical changes such as melting and freezing. Let’s think about what is happening in terms of energy transfers during melting and freezing. When you hold an ice cube in your hand, the solid water is converted to liquid water as it melts. How does this happen? How does your hand feel when this process is happening? Energy is being transferred from your hand to the ice cube to weaken the attraction between particles, allowing the solid to become a liquid. Now, what about the formation of an ice cube? Have you ever noticed that the outside of a freezer is often warm? Can you explain what is happening to make liquid water freeze into ice? Energy is being transferred from the liquid water out of the freezer making it warm to the touch.

These types of energy transfer are not only possible for physical processes but also chemical reactions. As this is the first introduction of chemical reactions so far in the course, let’s first look at some of the basics before we look more closely at energy transfers involved in chemical reactions.

Chemical reactions

We interact with all types of substance in our lives, from the fresh air we breathe, to the hot water we wash ourselves with and the soil we walk on. We are constantly interacting with elements, compounds and mixtures each and every day. Section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) shows that compounds consist of atoms of different elements chemically bonded together in a fixed ratio. But how do we form a compound starting with elements? How do we make new compounds from existing compounds?

Elements can undergo a chemical reaction with other elements to form a compound. Compounds can also undergo chemical reactions with other elements or compounds to form new compounds. A chemical reaction is a process where a rearrangement of atoms occurs forming new substance(s) with different properties. The initial substances used in a chemical reaction are called reactants and the new substances formed as a result of the chemical reaction are called products.

We can use a chemical equation to represent a chemical reaction. **Figure 4** shows a chemical equation for the formation of the compound carbon dioxide, CO_2, from its elements carbon, C, and oxygen, O_2.

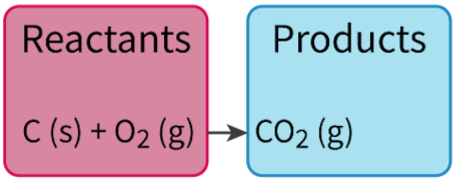

Figure 4. Chemical equation for the reaction forming carbon dioxide.

More information for figure 4

The image is a diagram depicting a chemical equation for the formation of carbon dioxide (CO2). On the left, a red box labeled "Reactants" contains the elements C (s) plus O2 (g). An arrow points from this box to a blue box on the right labeled "Products," which contains CO2 (g). The diagram illustrates the chemical reaction where solid carbon (C) reacts with gaseous oxygen (O2) to form gaseous carbon dioxide (CO2).

[Generated by AI]

Transfer of energy in chemical reactions

Think about chemical reactions you may have encountered such as wood burning in a campfire, plants photosynthesising, metabolising food or cooking an egg. What do all of these chemical reactions have in common? They all involve a transfer of energy. **All** chemical reactions involve a transfer of energy between the system and the surroundings, while total energy is conserved. In a chemical reaction, energy cannot be created or destroyed; instead, it is either converted from one form to another or transferred between the system and its surroundings. The system refers to the components involved in the chemical reaction and the surroundings encompasses everything outside the system. **Figure 5** shows the system and surroundings for the processes mentioned earlier.

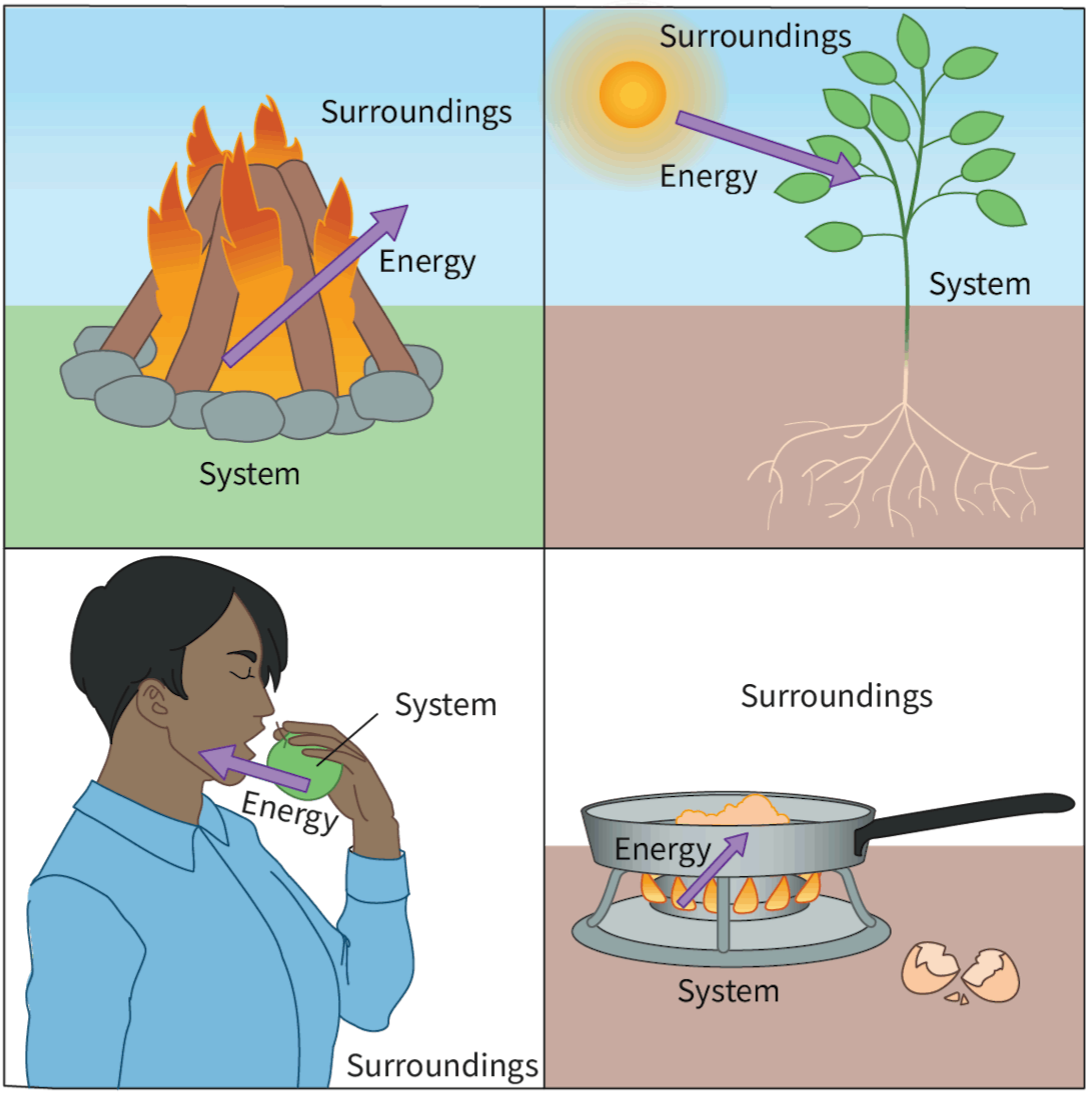

Figure 5. Common everyday chemical processes that involve a transfer of energy between system and surroundings.

More information for figure 5

This illustration shows four common chemical processes where energy transfer occurs between a system and its surroundings. The image is divided into four quadrants:

1. Top Left: A campfire with logs is depicted, flames rising into the air. An arrow labeled "Energy" points away from the "System" (the fire and logs) towards "Surroundings."

2. Top Right: A plant rooted in the ground under the sun is illustrated. An arrow labeled "Energy" points from the "Surroundings" (the sun) towards the "System" (the plant).

3. Bottom Left: A person is depicted eating, with an arrow labeled "Energy" pointing from a loaf of bread ("System") into their mouth, with "Surroundings" indicated around the person.

4. Bottom Right: A frying pan on a stove with a flame underneath is shown. An arrow labeled "Energy" points from the "Surroundings" (the heat source) into the "System" (the pan). This quadrant also shows broken eggshells, implying that something is being cooked in the pan.

This diagram illustrates how energy is transferred in different processes: combustion, photosynthesis, digestion, and cooking.

[Generated by AI]

In a reaction, such as the combustion reaction of methane (sometimes referred to as natural gas), methane reacts with oxygen to produce carbon dioxide and water via the following chemical reaction.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

The chemical potential energy stored in the bonds of the reactants is transformed into thermal energy and light energy during the combustion process. The total amount of energy is conserved.

Nature of Science

Aspect: Patterns and trends

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Here we have looked at how energy is able to change forms but the total amount of energy will be constant. This means that in a chemical reaction, there will be the same amount of total energy before and after a chemical reaction. This principle is so fundamental and important in science that it is called a law. This is the l**aw of conservation of energy** and is also known as the **first law of thermodynamics**.

If you study DP physics, you will discover that this law does not allow us to explain nuclear reactions, where there is an unexpected gain of total energy when some mass is lost. Albert Einstein proposed that what is happening is that the lost mass is instead converted into energy. This shows that even laws that appropriately describe much of our universe themselves have their limitations.

During a chemical reaction, energy may be released or absorbed in various forms such as heat, light or electrical energy. You have probably encountered each of these forms in different chemical reactions from processes such as combustion of wood in a campfire (heat), luciferin reacting in fireflies (light) and the oxidation and reduction reactions in a battery (electrical). These are shown in **Figures 6 to 8**.

Figure 6. Processes that release energy: fire.

Credit: Teemu Tretjakov, Getty Images

Figure 7. Processes that release energy: fireflies.

Credit: Ali Majdfar, Getty Images

Figure 8. Processes that release energy: batteries.

Credit: Peter Dazeley, Getty Images

Heat and temperature

Consider a chemical reaction taking place in a glass beaker that releases energy in the form of heat. What are some observations you would be able to make? How would energy be measured? Think about the equipment you have in your classroom laboratory: is there an 'energy meter'? If heat is being released by a reaction in the beaker and feels warm, what simple tool can you use to record quantitative data? When you have a fever, what instrument do you use to confirm that your body's temperature has increased as your body uses additional energy to fight an illness?

Temperature is a quantitative measurement often associated with heat. Section S1.1.3a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/temperature-and-kinetic-energy-id-43068/) mentions that temperature is a measure of the average kinetic energy of particles measured in units of K or °C. Heat is the energy transferred from a region of high temperature to a region of low temperature measured in units of J. These two terms are related but are different as illustrated in **Figure 9**. Temperature alone does not determine the amount of heat transfer. We will look at the other variables that are involved in heat transfer in section R1.1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpy-and-heat-transfers-id-43576/).

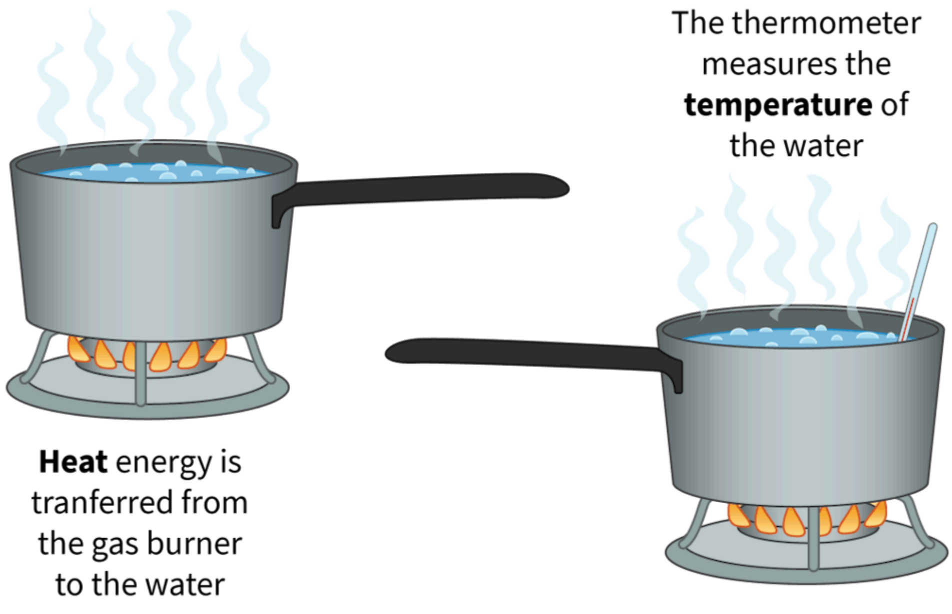

Figure 9. Comparison between heat and temperature.

More information for figure 9

The image depicts two scenarios involving pots on stoves to illustrate the difference between heat and temperature. On the left, a pot is placed on a gas burner with flames, indicating heat transfer from the burner to the water in the pot. Above this pot, steam is rising, and the text reads, "Heat energy is transferred from the gas burner to the water." On the right, another similar pot is on a gas burner with steam rising, but this time, a thermometer is placed in the water. The text next to this pot states, "The thermometer measures the temperature of the water." The image aims to show that while heat energy is transferred to increase the water's temperature, the thermometer measures the current temperature of the water.

[Generated by AI]

Theory of Knowledge

You have been provided with a lot of examples involving fire: an extreme release of heat that occurs in certain chemical reactions. The opposite of heat would be coldness, but what exactly is 'coldness'? Is it only the absence of heat, or is there something more?

There is historical medical evidence from many ancient cultures that viewed various maladies, such as cancer and infection, as a build-up of 'heat' in the body with the remedy to provide coldness in the form of cold water, ice baths or certain herbs designated as cold herbs, such as peppermint. Similarly, there were other illnesses that were thought to be caused by a build-up of coldness in the body, such as coughs: even the common cold! The treatment for these 'colds' was to provide heat such as warm clothing and herbs such as cinnamon that were thought to bring heat into the body.

Did isolated Indigenous societies intrinsically understand complex scientific processes, such as energy transfer?

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Thinking skills — Providing a reasoned argument to support conclusions
- **Time required to complete activity:** 30—45 minutes
- **Activity type:** Pair activity

Heat transfer and energy conservation

Aim: To investigate the different between temperature and heat, the difference between the system and surroundings and the law of conservation of energy

To perform this simple investigation, you will need the following materials:

- Two identical metal cans

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- Two thermometers
- Hot water
- Cold water
- Ice
- Stopwatch

Procedure:

- Fill each can up halfway with water: one can with hot water, the other with cold water.
- Measure and record the initial temperatures of water in each can.
- Place an ice cube into each can and start your stopwatch, stirring gently with the thermometer.
- Measure and record the temperatures of water in each can every minute for 10 minutes, or until the ice cubes melt completely.

Results and discussion:

- Represent your data in a well-organised data table.
- Calculate the change in temperature of the water for each can.
- Compare the changes in temperature for both cans and their water contents. Discuss which one had a greater change in temperature including an explanation of why.
- Explain what happened to the heat energy in each can during the experiment. Make sure you mention which part of the system gained heat and which part lost heat. Provide evidence for your reasoning.
- State the law of conservation of energy and discuss how it applies to this experiment.

R1.1 Measuring enthalpy changes

Endothermic and exothermic reactions

Learning outcomes

By the end of this section you should be able to:

- (R1.1.2) Identify the direction of energy transfer between system and surroundings for endothermic and exothermic reactions.
- (R1.1.2) Explain the temperature change (decrease or increase) that accompanies endothermic and exothermic reactions, respectively.
- (R1.1.3) Sketch and interpret energy profiles for endothermic and exothermic reactions.

It is common for people to associate chemical reactions with the release of energy. Usually this is because an increase in temperature of the surroundings is detected during many chemical reactions. Many reactions do, however, absorb energy. Arguably one of the most important chemical reactions on the planet requires the absorption of energy to convert carbon dioxide and water into carbohydrates and oxygen. What do you think this reaction might be?

Photosynthesis!

Photosynthesis: one of the most important reactions on the planet.

Credit: Sarayut Thaneerat, Getty Images

How might we describe reactions in terms of how energy is transferred? How might we expect the temperature to change as a result of energy transfer? How might we illustrate the energy changes of a chemical reaction graphically?

Direction of energy transfer

We can categorise chemical reactions in terms of their direction of energy transfer. Reactions that release energy to the surroundings are referred to as exothermic whereas reactions that absorb energy from the surroundings are referred to as endothermic. Energy transferred is typically in the form of heat but may also be electricity or light. **Figure 1** below illustrates the transfer of energy between system and surroundings for endothermic and exothermic processes.

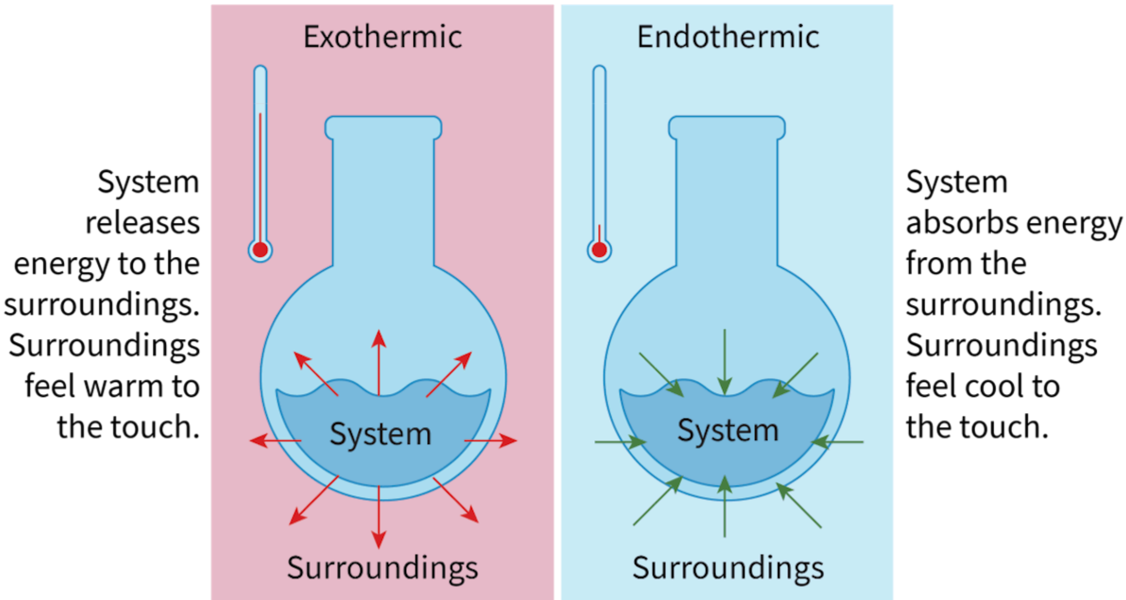

Figure 1. Transfer of energy during exothermic and endothermic processes.

More information for figure 1

The diagram is divided into two sections, with each section depicting a flask and a thermometer. The left section is labeled 'Exothermic' and shows a flask with arrows pointing outward from the system, indicating that energy is being released to the surroundings. The text beside it states, 'System releases energy to the surroundings. Surroundings feel warm to the touch.' The thermometer beside the flask shows a high temperature.

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The right section is labeled 'Endothermic' and shows a flask with arrows pointing inward to the system, indicating that energy is being absorbed from the surroundings. The text beside it reads, 'System absorbs energy from the surroundings. Surroundings feel cool to the touch.' The thermometer beside this flask shows a low temperature. Each section illustrates the direction of thermal energy transfer, affecting the temperature of the surroundings.

[Generated by AI]

Processes that are endothermic and exothermic

When you think of the physical changes of states covered in section S1.1.2a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/) can you identify which ones would be endothermic and which would be exothermic? Think about which processes involve an absorption of energy versus a release of energy. Try the drag and drop **Interactive 1** to classify the changes of state as endothermic or exothermic.

Interactive 1. Endothermic or Exothermic Processes

Can you think of any examples of chemical reactions that are always endothermic or exothermic? Think about the combustion engines in our cars or the process of plants making glucose from carbon dioxide and water. While many reactions will be exothermic or endothermic depending on the identity of the reactants, many types of reaction will always release or absorb energy. **Table 1** summarises some of these processes.

Table 1. Chemical and physical changes that are **always** endotherı

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	Exothermic		Er
	Type of reaction or process	Example	Type of reaction or process
Chemical change	Combustion	$CH_4(g) + O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$	Photosynthesis
Chemical change	Neutralisation	$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$	Thermal decomposition
Physical change	Freezing	$H_2O(l) \rightarrow H_2O(s)$	Melting
Physical change	Condensation	$H_2O(g) \rightarrow H_2O(l)$	Evaporation/boiling
Physical change	Deposition	$H_2O(g) \rightarrow H_2O(s)$	Sublimation

Effects on temperature

How do you expect the temperature to change during exothermic and endothermic reactions? When we measure the temperature of most reactions, are we measuring the temperature of the system or the surroundings? Let's consider an exothermic acid–base reaction in **Figure 2.**

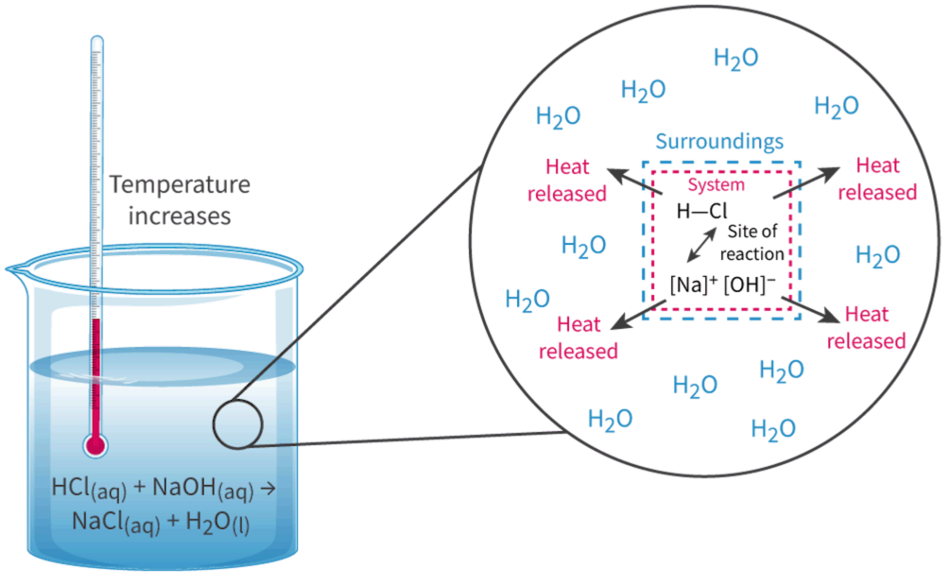

Figure 2. A close look at the system and surroundings during an exothermic acid—base reaction.

More information for figure 2

The image is a diagram illustrating an exothermic acid-base reaction. On the left side is a beaker filled with a liquid, depicted to represent the aqueous solution involving the reactants HCl(aq) and NaOH(aq). Above the liquid in the beaker, a thermometer shows the temperature increasing, indicative of the exothermic reaction.

To the right, a magnified view highlights the system and surroundings concept. Inside a box labeled "System," indicated by a dashed line, is where the reactants HCl and NaoH interact. Arrows point outward from the "System" to an area labeled "Surroundings," depicting the release of heat energy.

The surroundings are shown with multiple "H2O" labels, representing the solvent, water. This visualization emphasizes how heat from the reaction in the system is transferred to the water, increasing the temperature of the surroundings. The phrase "Heat released" accompanies each arrow pointing away from the "System." This setup serves to establish a clear distinction between the system and the surroundings in the context of chemical reactions.

[Generated by AI]

During most chemical reactions, when we measure the temperature we are often measuring the temperature of the surroundings. In **Figure 2**, we can see that an acid–base reaction involves an aqueous acid reacting with an aqueous base. Section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-

753301/book/elements-compounds-and-mixtures-id-43065/) shows that aqueous solutions involve a solute (HCl or NaOH in this example) dissolved in water as the solvent. When the reactants, HCl and NaOH, react, the site where these reactants react is the system. Everything around this site is a part of the surroundings. This means that the solvent, water, would act as the surroundings.

When the **exothermic** reaction between HCl and NaOH **releases energy** in the form of heat, the water molecules surrounding the site of reaction increase in kinetic energy. This results in an **increase in temperature** of the aqueous solution that we can measure using a thermometer.

What about endothermic processes? Have you ever felt an alcohol, such as that in hand sanitiser, on the surface of your skin? What does your skin feel like afterwards? The alcohol used in sanitiser alcohol is often used as an antiseptic to clean the skin. After use, the skin usually feels cool. This is due to the evaporation of the alcohol in the hand sanitiser!

As the alcohols used in hand sanitiser have a relatively low boiling point, the heat energy from your hands (surroundings) can be absorbed by the alcohol in the sanitiser (the system) leading to evaporation. The kinetic energy of molecules in your hands is slightly lowered due to this loss of heat energy resulting in a slightly lower temperature. This process is illustrated in **Figure 3**. The **endothermic** process of the evaporation of alcohol **absorbs energy** from the surroundings resulting in a **decrease in temperature** of the surroundings.

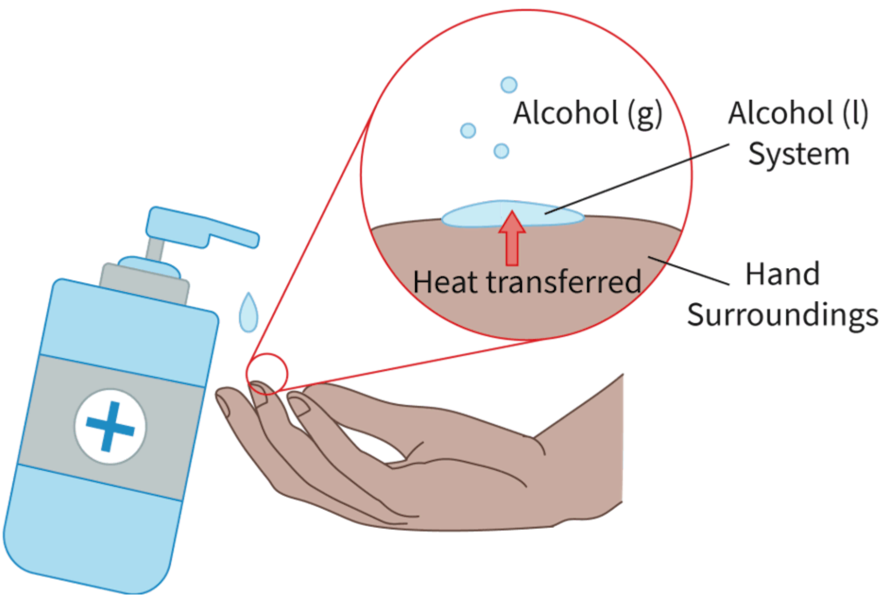

Figure 3. Hand sanitiser containing alcohol that easily evaporates in our hands.

More information for figure 3

The image is a diagram illustrating the evaporation process of hand sanitiser. It shows a hand with sanitiser being applied from a bottle. There is an enlarged view of the hand where a liquid droplet, labeled "Alcohol (l)", is shown on the palm, indicating the 'system'. Above it, there are smaller circles representing gaseous alcohol, labeled "Alcohol (g)", showing the evaporation process. An arrow labeled "Heat transferred" points from the hand to the droplet, representing the heat energy transfer from the hand to the alcohol, resulting in the evaporation. The hand is labeled as "Surroundings", and the concept of alcohol evaporating from liquid to gas is visually depicted with annotations.

[Generated by AI]

Energy profiles

Section R1.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/energy-transfer-in-chemical-reactions-id-43469/) considers different forms of energy that exist. Recall that one type of potential energy is chemical potential energy. Chemical potential energy is the energy stored in the chemical bonds between particles. The quantity of stored energy depends on the arrangement and type of bonding that exists between particles at the atomic level.

We can represent the chemical potential energy of products and reactants from endothermic and exothermic reactions graphically using an energy profile diagram. These energy profiles show the relative stability of reactants and products by considering the relative chemical potential energy each has. Look at the energy profile diagrams in **Figure 4**.

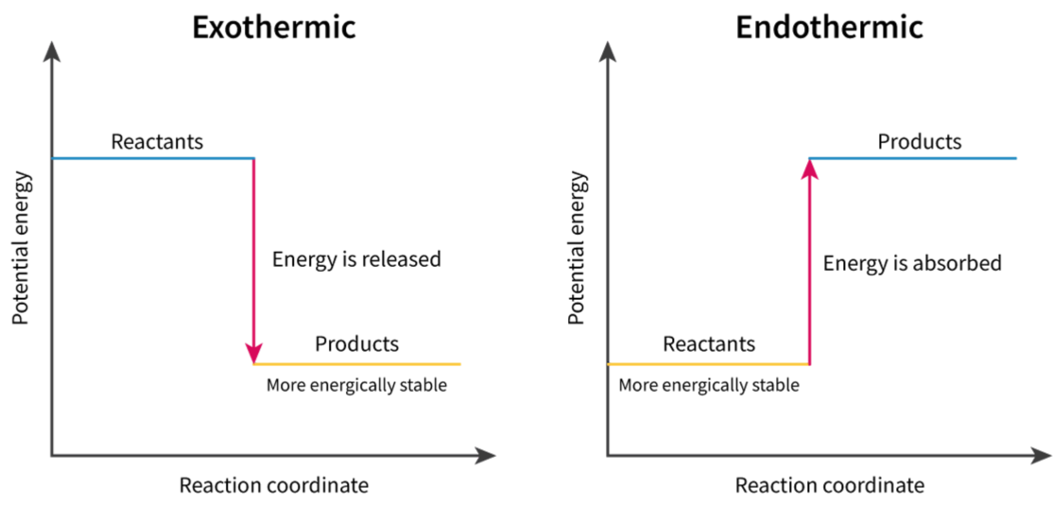

Figure 4. Energy profile diagrams for endothermic and exothermic reactions.

More information for figure 4

The image consists of two separate diagrams: an exothermic reaction on the left and an endothermic reaction on the right.

In the exothermic diagram: - The X-axis represents the progression of the reaction, while the Y-axis shows energy level. - The curve starts high, representing reactants with high potential energy, and drops sharply downwards, ending with products at a lower energy level. This indicates energy release.

In the endothermic diagram: - The X-axis also represents the reaction progress and the Y-axis shows energy. - This curve starts low, depicting reactants with lower potential energy, and rises to show that products have higher energy, representing absorbed energy.

Both diagrams illustrate the shifts in energy between reactants and products, showing how potential energy aligns with stability.

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You can see in **Figure 4** how exothermic processes show a release of energy with the products ending with a lower potential energy than the reactants. You can see here that stability and potential energy are interrelated. Similarly, in **Figure 4**, you can see how endothermic processes show an absorption of energy with the reactants having greater stability due to the lower potential energy. A lower potential energy leads to greater stability. You can think of it as being similar to a ladder. The lower on the ladder you stand the greater stability you have. The higher a ladder you stand the less stable you are.

Try the drag and drop in **Interactive 2** to match all missing terms to the correct places in the two energy profile diagrams.

Interactive 2. Energy profile diagrams.

More information for interactive 2

A drag-and-drop interactivity features two energy profile diagrams with missing terms. Users need to drag and drop the correct terms in the corresponding drop zones.
Both diagrams plot potential energy on the y-axis versus reaction coordinate on the x-axis.
In the first diagram, labeled drop zone, the reactants have a high potential energy and are converted into products with lower potential energy. Here, the energy is drop zone. Drop zones are more stable.
In the first diagram, labeled drop zone, the reactants have low potential energy and are converted into products with high potential energy. Here, the energy is drop zone. Drop zones are more stable.
The drag-and-drop options are as follows:
Released
Reactants
Exothermic
Products
Absorbed
Endothermic

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This interactivity helps users differentiate between exothermic and endothermic reactions based on energy profile diagrams.
Read below for the solution:
In the first diagram, labeled exothermic, the reactants have a high potential energy and are converted into products with lower potential energy. Here, the energy is released. Products are more stable.
In the second diagram, labeled endothermic, the reactants have a low potential energy and are converted into products with high potential energy. Here, the energy is absorbed. Reactants are more stable.

Activity

- **IB learner profile attribute:** Principled
- **Approaches to learning:**
  - Research skills — Using a single, standard method of referencing and citation
  - Thinking skills — Being curious about the natural world
- **Time required to complete activity:** 30—45 minutes
- **Activity type:** Individual activity

Research a physical process or chemical reaction with a notable energy transfer that is applicable to everyday life. Some examples you could choose include:

- sodium ethanoate (sodium acetate) heat packs
- hand warmers
- instant cold packs
- MREs (Meals, Ready to Eat or ration packs)
- cellular respiration
- formation of snow
- making ice cream.

Create an infographic (using a digital tool such as Canva (https://www.canva.com/)) showing what you have learned. Your infographic should include the following:

- Identification of the chemical or physical process that is occurring.
- Classification of the process as either endothermic or exothermic.
- Diagram labelling system, surroundings and direction of energy transfer.
- Description of expected temperature changes.
- Energy profile diagram.

Kog

- Why is energy transfer important in this real-world application.
- 3—5 interesting facts.
- Proper citations for any references used.

Enthalpy and heat transfers

Learning outcomes

By the end of this section you should be able to:

- Deduce whether a reaction is endothermic or exothermic from a thermochemical equation.
- Analyse data from calorimetry experiments to determine the change in temperature from a chemical reaction.
- Use changes in temperature to calculate the thermal energy, *Q*, and the enthalpy change of a reaction, Δ*H*.

Biodiesel is a fuel that is produced from waste vegetable oil or waste animal fats. This is considered a sustainable fuel source because the waste oil takes the place of fossil fuel. Methyl palmitate is a substance derived from vegetable oils that can undergo an exothermic combustion reaction. The release of energy from this reaction can be used for transportation fuels, heating or electricity generation. The combustion reaction of methyl palmitate is:

$$C_{17}H_{34}O_2(l) + \frac{49}{2}O_2(g) \rightarrow 17CO_2(g) + 17H_2O(g)$$

How might we quantify the energy transferred for specific chemical reactions? What are some ways in which we could measure the energy released in a chemical reaction such as the combustion of methyl palmitate?

Creativity, activity, service

Strand: Service
Learning outcome: Demonstrate engagement with issues of global significance

The combustion of fossil fuels is thought to contribute to global warming and climate change that could have serious consequences for the environment. One possible solution is to increase the use of renewable energy sources (https://www.un.org/en/climatechange/what-is-renewable-energy) such as biofuels and solar cells. As an IB student, can you think of any ways to raise awareness of such issues in your school or local community? One possible CAS experience could be to research the use of renewable energy sources and suggest ways in which their usage could be increased.

Enthalpy and standard enthalpy change of reaction

The amount of heat that a system has is known as its enthalpy (H). The enthalpy of a system is not easily measured but the changes in enthalpy (ΔH) is something we can measure as heat is transferred between system and surroundings. These changes in enthalpy are positive or negative depending whether a reaction absorbs or releases heat.

Exothermic reactions, involve heat being released, indicating a negative change in enthalpy. Endothermic reactions involve heat being absorbed, resulting in a positive change in enthalpy.

Standard enthalpy change of reaction $(\Delta H^{\ominus})$ refers to the heat transferred at constant pressure and under standard conditions and states. Enthalpy change (ΔH) or standard enthalpy change $(\Delta H^{\ominus})$ can be included with a chemical reaction providing information about heat transfer for that process. When enthalpy is included beside a chemical reaction, it is known as a thermochemical equation. Next are some examples of thermochemical equations for the combustion of methane and the decomposition of calcium carbonate. **Figure 1** shows an energy profile diagram for these reactions.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \qquad \Delta H^{\ominus} = -891\ \text{kJ mol}^{-1}$$

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g) \qquad \Delta H^{\ominus} = +180\ \text{kJ mol}^{-1}$$

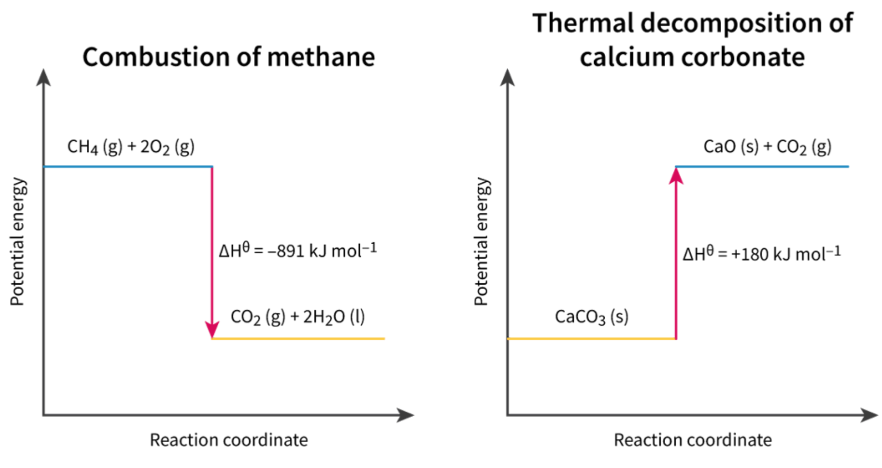

Figure 1. Energy profile diagrams for the combustion of methane and the thermal decomposition of calcium carbonate.

More information for figure 1

Notice how the units of enthalpy are in units of kJ mol^{-1}. This indicates the energy change associated with one mole of a chemical species identified in the chemical reaction. For example, 891 kJ of energy are released for every 1 mole of methane that has undergone combustion and 180 kJ of energy are absorbed for every 1 mole of calcium carbonate that has undergone decomposition.

Thermal energy or heat

Section R1.1.2–3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/endothermic-and-exothermic-reactions-id-45282/) shows how heat and temperature, while distinct, are related to one another. Energy changes cannot be measured directly in J or kJ, but instead are monitored by temperature changes and then related to thermal energy via the following mathematical relationship shown in **Table 1**.

Table 1. Equations for thermal energy.

Equation	Symbol	Units
$Q = mc\Delta T$	Q = thermal energy (heat absorbed or released)	joules, J
	M = mass (of the species you are measuring the change in temperature of)	grams, g
	c = specific heat capacity (specific to the substance undergoing a change in temperature)	joules per gram [unit Kelvin, J g^{-1}
	ΔT = change in temperature	Celsius or Kelvi[°C or K

Study skills

You may wonder why the units of change in temperature are either in °C or K. This is because the conversion between the two units involves a difference of 273.15. For example, if a sample has an initial temperature of 0 °C that is 273.15 K and ends with a final temperature of 25°C that is 298.15 K, the difference in temperature would be 25 °C or 25 K as shown below. This **only** applies to a difference in temperature and not the temperature measurement itself.

ΔT in Celsius	ΔT in Kelvin
$\Delta T = 25\,°\text{C} - 0\,°\text{C}$ $\Delta T = 25\,°\text{C}$	$\Delta T = 298.15\text{ K} - 273.15\text{ K}$ $\Delta T = 25.00\text{ K}$

The specific heat capacity of a substance is the quantity of energy needed to raise 1 g of a substance by 1 K (or 1 $^{\text{O}}$C). It is unique to the identity of the substance. **Table 2** shows the specific heat capacity for some common substances.

Table 2. Specific heat capacities for some common substances.

Substance	Specific heat capacity (J g^{-1} K^{-1})
Water	4.18
Copper	0.39
Graphite	0.71
Steel	0.47

You can see in **Table 2** that water has a very high specific heat capacity in comparison to the other materials listed. This means it takes a lot of energy to increase the temperature of water and will take a long time to cool down. Conversely, copper does not take as much energy to increase the temperature and will cool down quickly.

Worked example 1

A 150.0 g block of pure copper is heated so that the temperature increases from 18.0°C to 75.0°C. Calculate the heat energy change in kJ if the specific heat capacity of copper is 3.86×10^{-1} J g^{-1} K^{-1}.

To calculate the heat energy change we can use the equation:

$$Q = mc\Delta T$$

Substituting the values for the variables given in the question we have:

$$Q = (150.0\ \text{g})(3.86 \times 10^{-1}\text{J g}^{-1}\ \text{K}^{-1})(75.0\,^{\circ}\text{C} - 18.0\,^{\circ}\text{C})$$

$$Q = 3300.3\ \text{J}$$

The final step would be to convert to kJ by dividing by 1000 and reporting to the appropriate number of significant figures.

$$Q = \frac{3300.3}{1000}\ \text{kJ}$$

Therefore, the heat energy change would be.

$$Q = 3.30\ \text{kJ}$$

Calorimetry to determine the enthalpy change of a reaction

Now we know how to relate temperature changes to energy change from a chemical reaction, how can we physically measure these changes in a laboratory setting? Calorimetry is a technique used to measure the heat transfer during a physical or chemical process. A calorimeter is the apparatus used to measure temperature changes to calculate the enthalpy for a reaction. Something as simple as a Styrofoam coffee cup can be sufficient to act as a calorimeter in certain reactions. **Interactive 1** below shows some examples of different calorimeters along with details about the situation you are likely to use each for.

Interactive 1. Types of calorimeter and when to use them.

More information for interactive 1

Practical skills

Tools 1: Experimental techniques — Measuring variables

Inquiry 1: Exploring and designing — Exploring

You might have noticed in **Interactive 1** that all the calorimeters use water. Why do you think that is? Water has a relatively high heat capacity relative to other substances. This means large amounts of heat produced will result in relatively small temperature changes in water compared to other substances that have low heat capacities. This makes water a practical and safe substance to use for measurement as it will not change state easily or reach temperatures beyond the range of a thermometer.

Additionally, water has a density of 1.00 g cm^{-3} over a large range of temperatures (ranging from 1.00 g dm^{-3} at 0 °C to 0.96 g cm^{-3} at 100 °C). This allows us to easily relate the volume of the water in the calorimeter to the mass of water changing in temperature.

We can use a calorimeter to measure temperature changes from chemical reactions and calculate the thermal energy transferred in the reaction. To make these quantities specific to a chemical reaction, we can use the thermal energy transferred (Q) to calculate the change in enthalpy for a reaction (ΔH).

You may have noticed that the units of ΔH are kJ mol^{-1}; this means we need to involve the number of moles involved in a chemical reaction to calculate enthalpy. Additionally, calorimetry involves measuring the temperature of the surroundings, this means that the thermal energy (Q) and the change in enthalpy (ΔH) will be opposite in sign to one another. To account for this, we can calculate the change in enthalpy ΔH using the equation in **Table 3**.

Table 3. Equation for change in enthalpy.

Equation	Symbols	Units
$\Delta H = -\frac{Q}{n}$	ΔH = change in enthalpy	kilojoules per r kJ mol^{-1}
	Q = thermal energy (heat absorbed or released)	joules, J
	n = number of moles of an identified reactant or product in the chemical reaction	moles

Try the **Worked example 2** to calculate the enthalpy of combustion for ethanol.

Worked example 2

A copper calorimeter containing 100.0 g of water is used to measure the enthalpy of combustion of ethanol. Water in the copper calorimeter has an initial temperature of 21.0 °C and a final temperature of 67.0 °C. The burner containing ethanol had an initial mass of 25.83 g and a final mass of 21.75 g. Calculate the enthalpy of combustion of ethanol in kJ mol^{-1}.

To calculate the enthalpy of combustion of ethanol we need to first calculate the thermal energy that was transferred from the reaction to the water in the calorimeter. To do this, we use:

$$Q = mc\Delta T$$

As water is the entity changing in temperature, we must use the mass and specific heat capacity of water in this equation:

$$Q = (100.0)(4.18)(67 - 21)$$

$$Q = 19\,228 \text{ J}$$

To calculate the enthalpy change for combustion of ethanol we need to use the equation:

$$\Delta H = -\frac{Q}{n}$$

Since ethanol is the identified species of interest here, we will need to calculate the number of moles of ethanol. First, we need to determine the mass of ethanol that reacted. We can do this by taking the difference of the final and initial mass of the ethanol burner.

$$m = 25.83 - 21.75$$

$$m = 4.08 \text{ g}$$

The chemical formula of ethanol is CH_3CH_2OH, so the molar mass is 46.08 g mol^{-1}.

$$n = \frac{m}{M}$$

$$n = \frac{4.08}{46.08}$$

$$n = 0.0885 \text{ mol}$$

Now we can calculate the change in enthalpy for the combustion of ethanol:

$$\Delta H = -\frac{Q}{n}$$

$$\Delta H = -\frac{19\ 228 \text{ J}}{0.0885 \text{ mol}}$$

$$\Delta H = -217\,163 \text{ J mol}^{-1}$$

To convert to kJ

$$\Delta H = -\frac{217\,163}{1000} \text{ kJ mol}^{-1}$$

$$\Delta H = -217 \text{ kJ mol}^{-1}$$

Therefore, the change in enthalpy for the combustion of ethanol is –217 kJ mol^{-1}.

Study skills

When using the equation $Q = mc\Delta T$, a common error made is to use the incorrect mass for the variable m. You should always ask yourself, what is the substance that is changing in temperature? If it is water, then the mass and specific heat capacity should be of water. The variables m, c and ΔT should always be for the same substance.

Determining the enthalpy change involving aqueous reactants

Calculations can sometimes become tricky when determining the change in enthalpy for reactions involving aqueous solutions. Let's consider a chemical reaction involving aqueous copper sulfate and solid zinc as an example.

$$CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$$

The reaction is exothermic and the release of energy can be monitored by recording the temperature increase over time. But when calculating the thermal energy released, what mass and specific heat capacity would we use in the equation $Q = mc\Delta T$? Because the temperature change measured would be of the aqueous solution, the mass and specific heat capacity would also need to be that of an aqueous solution.

As the reactants are being consumed and the products are forming, their masses are changing throughout the reactions and therefore cannot be used in the $Q = mc\Delta T$ equation. The energy changes instead are monitored by the water solvent.

As water has a density of 1.00 g cm^{-3}, we can easily convert between volumes of water/aqueous solutions and masses when solving for thermal energy. Let's try applying some of these assumptions below to calculate the change in enthalpy for the reaction between zinc and copper(II) sulfate in the **Worked example 3**.

Worked example 3

Calculate the enthalpy change of reaction in kJ mol^{-1} for the reaction of

$$CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$$

Given that excess solid zinc reacted with 150 cm^3 of 1.0 mol dm^{-3} copper(II) sulfate resulting in a temperature rise of 43 °C.

To calculate the enthalpy of reaction, we need to make use of the following two equations:

$$\Delta H = -\frac{Q}{n} \text{ and } Q = mc\Delta T$$

The aqueous solution would experience the measured temperature rise of 43 °C, therefore the mass and specific heat capacity of the aqueous solution can be assumed to be that of water.

As the volume of the aqueous solution is 150 cm^3 and water has a density of 1.00 g cm^{-3}, this would mean the mass of the solution would be 150 g.

Therefore, the thermal energy can be calculated as

$$Q = mc\Delta T$$

$$Q = (150)(4.18)(43)$$

$$Q = 26\,961 \text{ J}$$

To find the change in enthalpy we will need to calculate the number of moles. The substance indicated in the question is copper(II) sulfate (because zinc is in excess) so we can determine the number of moles of $CuSO_4$ using the concentration and volume ensuring the volume is used in dm^3.

$$n = CV$$

$$n = (1.0)\left(\frac{150}{1000}\right)$$

$$n = 0.150 \text{ mol}$$

Now we have all information needed to calculate the change in enthalpy of the reaction:

$$\Delta H = -\frac{Q}{n}$$

$$\Delta H = -\frac{26\,961}{0.150}$$

$$\Delta H = -179\,740 \text{ J mol}^{-1}$$

Finally, we convert to kJ mol^{-1} and report the result to an appropriate number of significant figures

$$\Delta H = -\frac{179\,740}{1000}$$

$$\Delta H = -180\,\text{kJ mol}^{-1}$$

Accounting for heat loss during data processing

When you consider using a calorimeter to determine the change in enthalpy for a reaction, what do you think the largest source of error would be? The system might be closed but it is not isolated; therefore, it is always 'connected' to the earth in some way, making heat loss unavoidable. This heat loss results in smaller changes in temperature than we would expect theoretically. This means that experimentally, we would record a final temperature lower than theoretically possible for exothermic reactions and a final temperature higher than theoretically possible for endothermic reactions.

Can you think of any ways we can reduce heat loss experimentally? Insulation of calorimeters is one way to reduce heat loss. This can be done by surrounding the calorimeter with insulating materials, increasing the thickness of these materials and minimising contact with the air.

Can you think of any ways we might be able to process data to account for this heat loss? Let's look at how the temperature changes during the reaction of excess solid zinc reacting with a 150 cm^3 solution of 1.0 mol dm^{-3} copper(II) sulfate in **Figure 2**.

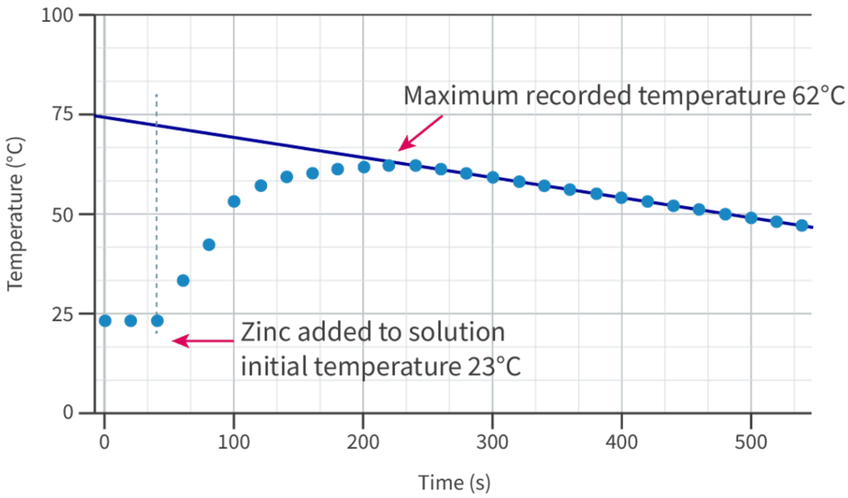

Figure 2. Temperature changes during the reaction of solid zinc with aqueous copper(II) sulfate.

More information for figure 2

In **Figure 2**, we can see how it takes time before we are able to record the maximum temperature of this exothermic reaction. During this time, heat would be constantly lost to the air, calorimeter and unreacted zinc. Ideally, heat would be lost to these factors at the same rate. This would mean that heat would be lost linearly while the solution cools. We can take advantage of the linear rate of cooling to estimate the theoretical maximum temperature. **Figure 3** shows how this might be done.

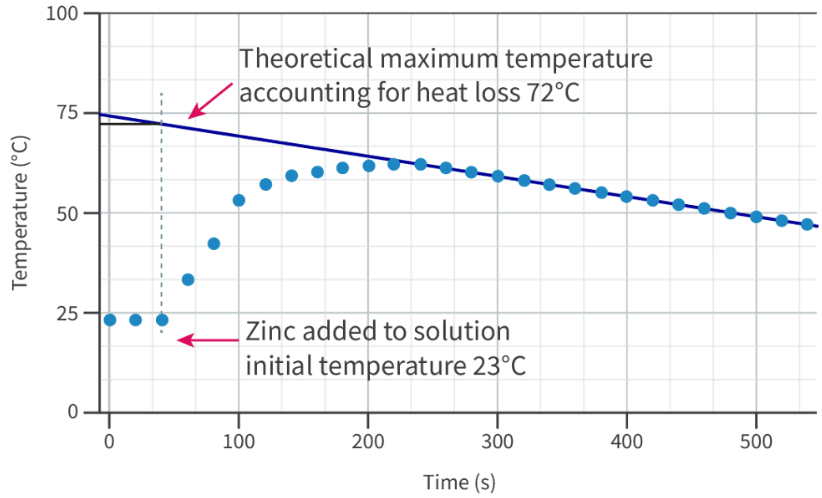

Figure 3. Extrapolation to account for heat loss to determine a theoretical final temperature.

More information for figure 3

Figure 3 shows how by extrapolating the line representing the rate of cooling back to the point where the reaction started can provide a good estimate of the theoretical final temperature. We now have all information needed to solve for this change in enthalpy of reaction. Try it yourself first, and then check with the solutions in **Worked example 4**.

Worked example 4

Calculate the enthalpy change of reaction for the reaction of

$$CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$$

Given that excess solid zinc reacted with 150 cm^3 of 1.0 mol dm^{-3} copper(II) sulfate and showed the temperature changes shown in **Figures 2 and 3.**

To calculate the enthalpy of reaction we need to make use of the following two equations.

$$\Delta H = -\frac{Q}{n} \text{ and } Q = mc\Delta T$$

The extrapolated value for the maximum theoretical temperature would need to be used when calculating the temperature change:

$$\Delta T = 72 - 23$$

$$\Delta T = 49\,^{\circ}\mathrm{C}$$

The aqueous solution would experience the measured temperature rise of 49 °C, therefore the mass and specific heat capacity of the aqueous solution can be assumed to be that of water.

As the volume of the aqueous solution is 150 cm^3 and water has a density of 1.00 g cm^{-3}, this would mean the mass of the solution would be 150 g.

Therefore, the thermal energy can be calculated as

$$Q = mc\Delta T$$

$$Q = (150)(4.18)(49)$$

$$Q = 30723 \text{ J}$$

To find the change in enthalpy, we will need to calculate the number of moles. The substance indicated in the question is copper(II) sulfate (because zinc is in excess) so we can determine the number of moles of $CuSO_4$ using the concentration and volume ensuring the volume is used in dm^3.

$$n = CV$$

$$n = (1.0)\left(\frac{150}{1000}\right)$$

$$n = 0.150 \text{ mol}$$

Now we have all information needed to calculate the change in enthalpy of the reaction:

$$\Delta H = -\frac{Q}{n}$$

$$\Delta H = -\frac{30723}{0.150}$$

$$\Delta H = -204\,820 \text{ J mol}^{-1}$$

Finally, convert to kJ mol^{-1} and report to an appropriate number of significant figures:

$$\Delta H = -\frac{204\,820}{1000}$$

$$\Delta H = -205 \text{ kJ mol}^{-1}$$

Aspect: Measurement

Producing knowledge in thermochemistry is possible because of assumptions made. These assumptions also limit the validity of the knowledge produced. There are benefits and limitations associated with making assumptions and these must be evaluated and justified for any scientific investigation.

The following includes a summary of assumptions made in determining the change in enthalpy of reaction for zinc and copper(II) sulfate:

- All energy released from the reaction is transferred to the water solvent.
- The solution is dilute therefore negligible volume from solute.
- The excess zinc powder and produced copper solid do not absorb any of the heat released from the reaction.
- Solution loses heat to the surroundings linearly.
- Water has a density of 1.00 g cm^{-3}.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Communication skills
    - Presenting data appropriately
    - Using symbols and communication conventions consistently
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual activity

When hydrochloric acid, HCl, reacts with sodium carbonate an exothermic reaction occurs.

$$2HCl(aq) + Na_2CO_3(s) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$$

When hydrochloric acid, HCl, reacts with sodium hydrogencarbonate an endothermic reaction occurs.

$$HCl(aq) + NaHCO_3(s) \rightarrow NaCl(aq) + H_2O(l) + CO_2(g)$$

The temperature data can be monitored throughout each of these reactions using a calorimeter. The following table shows an example of experimental temperature data collected from a student's experiments. Each experiment involves the reaction of excess carbonate with 75.0 cm^3 of 1.50 mol dm^{-3} HCl(aq).

Time (s)	Temperature (°C) for reaction of sodium carbonate with hydrochloric acid	h
0	22.0	
20	22.0	
40	22.0	
60	22.0	
80	25.0	
100	28.0	
120	32.0	
140	35.0	
160	37.0	
180	38.0	
200	39.0	
220	40.0	
240	39.0	
260	39.0	
280	39.0	
300	38.0	
320	37.0	
340	36.0	
360	35.0	

Time (s)	Temperature (°C) for reaction of sodium carbonate with hydrochloric acid	h
380	34.0	
400	33.0	
420	32.0	

For each reaction, complete the following:

- Plot temperature data and carry out an extrapolation to determine the theoretical temperature accounting for heat loss.
- Calculate the change in enthalpy.
- Sketch an energy profile diagram.
- Describe what the system and surroundings are in these experiments.

R1.1 Measuring enthalpy changes

The big picture

Guiding question(s)

What can be deduced from the temperature change that accompanies chemical or physical change?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

What a great time to be born! What a great time to be alive! Because that generation gets to essentially completely change the world.

Quote from Paul Hawken (https://paulhawken.com/) an environmentalist, entrepreneur, author and activist from the USA

It might not always feel like it, but people in modern societies have longer, healthier lives with more opportunities to work towards happiness than any era ever before. Many developments were essential for us to get here, like the invention of writing, farming and even spoken language, but earlier and perhaps even more important as any of these advancements was the ability to recognise the interconnectedness between changes in temperature and chemical changes of edible and material goods.

Indigenous people have long used energy transfer to prepare foods such as fish through methods of heating directly in a fire, drying in the sun and steaming in a ground oven. Māori *hāngī* refers to the traditional method of cooking food on top of heated rocks buried underground by the Māori people in New Zealand. Long before modern science was developed, it was known, intuitively, that the interactions with heat transformed the fish into something safe to eat for sustenance. This Indigenous knowledge has been essential in the evolution of our species. Some of these methods of fish preparation are shown in **Interactive 1**.

Interactive 1. Indigenous Methods to Prepare Foods.

More information for interactive 1

The ability of humans to be able to use energy transfers to prepare food, create materials for infrastructure, design diagnostic tests to monitor health and create transportation methods to allow travel all over the world are some examples of reasons today's societies live longer, healthier lives.

Prior learning

Before you study this subtopic, make sure that you understand the following:

- Convert between values in the Celsius and Kelvin scales (see section S1.1.3a (/study/app/preview-p/sid-426-cid-753301/book/temperature-and-kinetic-energy-id-43068/)).
- Solve problems involving the relationships between the number of particles, the amount of substance in moles and the mass in grams (see section S1.4.3 (/study/app/preview-p/sid-426-cid-753301/book/molar-mass-m-id-44262/)).
- Solve problems involving the molar concentration, amount of solute and volume of solution (see section S1.4.5a (/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/)).

Practical skills

Once you have completed this subtopic, apply your knowledge of energy transfers by going to Practical 5a: Determining enthalpy changes in aqueous solutions (/study/app/preview-p/sid-426-cid-753301/book/determining-enthalpy-changes-in-aqueous-solutions-5a-id-46719/) and Practical 5b: Determining enthalpy changes for combustion reactions (/study/app/preview-p/sid-426-cid-753301/book/determining-enthalpy-changes-for-combustion-reactions-5b-id-46720/).